

MAAHIR RAJESH GARG

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Education

National University of Singapore

Aug 2023 – May 2027 (Expected)

B.Comp. (Honours), Computer Science; Second Major in Quantitative Finance | GPA 4.71 / 5.00

Singapore

- Minor in Data Engineering; AI Specialisation (Distinction). Coursework: Algorithms, Advanced Mathematics in Quantitative Finance, Probability & Statistics, Regression, Machine Learning, Multivariable Calculus.

Technical Skills

Quantitative: probability, statistics, regression, optimisation, stochastic processes, hypothesis testing

AI & ML: PyTorch, Hugging Face, scikit-learn, NumPy, pandas, PEFT/LoRA/BitFit, quantisation, pruning

Programming: Python, C/C++ , Java, SQL, R, Swift, JavaScript/TypeScript, Bash, LaTeX

Systems & Data: FastAPI, Pydantic, pytest, agentic LLMs, Arize, Spark, Snowflake, PostgreSQL, Supabase, Docker, AWS, Git

Experience

AI Centre for Educational Technologies, Team Koditsu

Aug 2026 – Present

AI Engineer | AI/ML, Sequence Modelling, LLMs

Singapore

- Developing model-based cheating-risk signals for Koditsu's coding-assessment platform from character-level editor interactions, moving beyond fixed event rules.
- Investigating sequence-model and LLM approaches that combine event streams with code snapshots to produce probabilistic scores for human review.

GIC

Jan 2026 – Jul 2026

AI Engineer | Python, Agentic LLMs, Arize, Chainlit, Snowflake, Docker/AWS

Singapore

- Built an internal **agentic CLI and Chainlit assistant** over proprietary LLMs in a sandboxed Docker/AWS environment, with **8 custom tools**, multi-agent orchestration, and skill-routing for an **11-person team**.
- Designed a Snowflake/SharePoint agentic pipeline that generated audit-ready reports from plain-English prompts, automated **2 previously manual data sources**, and cut evidence retrieval from **hours to near-instant**.
- Automated **50+ audit tests** and added step-level guardrails through custom hooks, reducing manual oversight.
- Built custom evaluations and **Arize tracing** over model/tool execution, making runs inspectable for audit and debugging.

Interactive 3D Lab, in collaboration with Apple

May 2025 – Dec 2025

Swift Developer, Apple Vision Pro | Swift, RealityKit, visionOS, ARKit

Singapore

- Co-developed with Apple a Vision Pro/iPhone tracking surface at **89% cross-device accuracy** for stroke-rehabilitation hand-eye assessment.
- Built the visionOS spatial UI, hand/gaze tracking, and iPhone companion app at **30 fps** and **sub-ms latency**; co-author on a paper in preparation and co-inventor on a patent application in progress.

Accelerice, NUS Overseas Colleges

May 2024 – Aug 2024

Data Engineer | Python, APIs, Databases, Web Performance

Jakarta, Indonesia

- Built Python/API dashboards and backend databases that improved operations-team data access by **40%**.
- Raised search rankings **30%** and organic traffic **20%** while cutting page-load time **25%**.

Projects

Trader's Edge (Final-Year Project) | AI, Strategy Backtesting, Team Repository

May 2026 – Present

- Building a team FYP, advised by Anand Bhojan, that helps students specify and backtest trading strategies in an interactive learning environment.
- Contributing to a working MVP that pairs plain-English strategy specification and historical-data backtesting; public demo requires Google sign-in.

Model Compression for BERT Question Answering | PyTorch, Quantisation, Pruning

Nov 2025

- Reduced BERT QA model size by **69%**, from 440MB to 128MB, using post-training quantisation while limiting F1 loss to **0.5%**.
- Implemented custom mask-enforced pruning to reach **54.7% sparsity** and profiled the efficiency-versus-generalisation trade-off.

Prediction-Market Arbitrage & Research Engine | Python, FastAPI, Pydantic, pytest

Feb 2026 – Jul 2026

- Built a **read-only FastAPI engine** over Polymarket/Kalshi, modelling human-confirmed mispricings after fees, liquidity, and hedge-equal fills.
- Validated matching, calibration, fees, and API failure modes with **36+ tests** and live-endpoint checks.

Honours

JEE top 0.5% of 1.2M; **National Topper in Mathematics (97.2% overall)**; NUS Tutor Honour List; AI Specialisation with Distinction.